



KAMBALA

Student Number: _____

**Yearly Examination
September 2020**

Preliminary Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Board-approved calculators may be used
- A Reference Sheet is provided for use in this task
- Answer questions 1 – 10 on the answer sheet provided
- Answer questions 11 through 26 on the paper provided
- For questions 11 through 26, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II

60 marks

- Attempt Questions 11 through 26
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

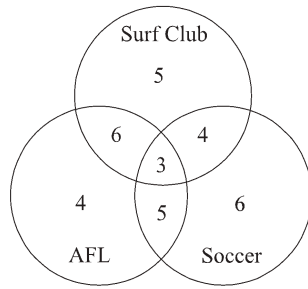
Attempt questions 1-10.

Allow about 15 minutes for this section.

Use the answer sheet for Questions 1 – 10.

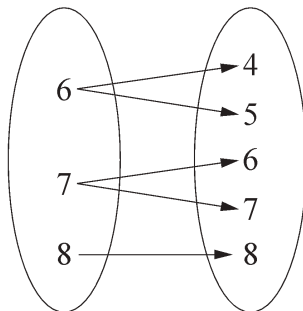
1. What is the value of $\sqrt{48} - \sqrt{12}$?
 - (A) 2
 - (B) $2\sqrt{3}$
 - (C) $2\sqrt{6}$
 - (D) 6
2. What are the x intercepts of the parabola $y = x^2 - 2x - 3$?
 - (A) -3 and -1
 - (B) -3 and 1
 - (C) -1 and 3
 - (D) 1 and 3
3. What are the solutions of the equation $|2x + 1| - 3 = 0$?
 - (A) $x = 2$ or $x = -1$
 - (B) $x = -2$ or $x = -1$
 - (C) $x = -2$ or $x = 1$
 - (D) $x = 2$ or $x = 1$
4. What is the perimeter of a sector of a circle of radius 6cm which contains a sector angle of $\frac{\pi}{6}$ radians at the centre of the circle?
 - (A) π cm
 - (B) 3π cm
 - (C) $(12 + \pi)$ cm
 - (D) $(12 + 3\pi)$ cm

5. In a classroom, students are asked what sports club they are members of and the results are shown in the Venn diagram.



A student who is a member of the soccer club is chosen at random. What is the probability that he/she is also a member of a surf club?

- (A) $\frac{2}{5}$
- (B) $\frac{4}{11}$
- (C) $\frac{2}{9}$
- (D) $\frac{7}{18}$
6. What type of relation is this?



- (A) Many-to-many
- (B) One-to-many
- (C) One-to-one
- (D) Many-to-one

7. If $y = \sqrt{2x + 1}$ what is the expression for $\frac{dy}{dx}$?

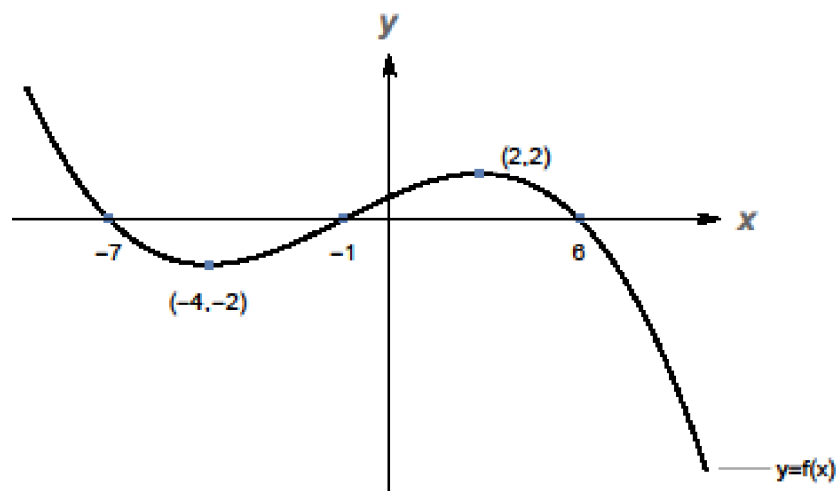
(A) $\frac{dy}{dx} = 2\sqrt{2x + 1}$

(B) $\frac{dy}{dx} = \frac{1}{2\sqrt{(2x+1)}}$

(C) $\frac{dy}{dx} = \frac{1}{\sqrt{(2x+1)}}$

(D) $\frac{dy}{dx} = \frac{2}{\sqrt{(2x+1)}}$

8.



For the graph $y = f(x)$ shown above, $f'(x)$ is negative when

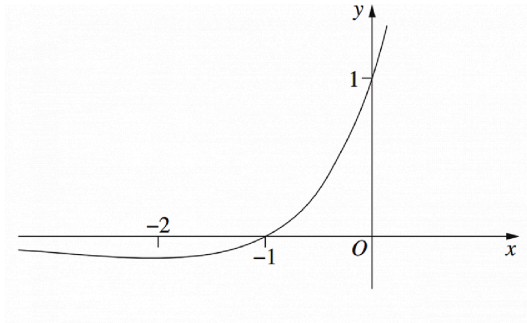
(A) $-7 < x < -1$ or $x > 6$

(B) $x < -4$ or $x > 2$

(C) $-4 < x < 2$

(D) $-2 < x < 2$

9. The diagram shows the graph of $y = e^x(1 + x)$ below.



How many solutions to $y = e^x(1 + x) = 1 - x^2$?

- (A) 0
(B) 1
(C) 2
(D) 3
10. If $a = e^x$ what is the correct expression for $\log_e \frac{1}{a^2}$?

- (A) $-x^2$
(B) $-2x$
(C) $2x$
(D) x^2

Section II

60 marks

Attempt all questions.

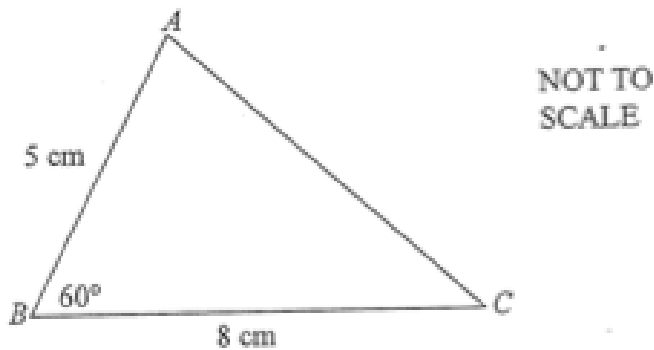
Allow about 1 hour and 45 minutes for this section.

Answer the questions in the spaces provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the back of the examination paper.

Question 11 (3 marks)



ABC is a triangle in which $AB = 5\text{cm}$, $BC = 8\text{cm}$ and $\angle ABC = 60^\circ$

(a) Find the length of AC.

2

(b) Find in simplest exact form the area of $\triangle ABC$

1

Question 12 (2 marks)

(a) Solve for x

$$\frac{x-2}{4} + \frac{7-x}{6} = 1$$

2

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Question 13 (2 marks)

2 A(− 2, 1), B(2, − 1) and C (4, 3) are the vertices of a triangle. Show that $AB \perp BC$.

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Question 14 (2 marks)

A fair die is rolled once. A is the event *the score is even* and B is the event *a multiple of 3*. Explain whether events A and B are independent. **2**

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Question 15 (3 marks)

- (a) If $f(x) = 4 - x^2$ and $g(x) = \sqrt{x}$ find the composite function $f(g(x))$. **1**

- (b) Sketch the graph of this composite function showing the x and y intercepts. **2**

Question 16 (5 marks)

- (a) Differentiate **1**

$$x^4 + 2x^2 + 10$$

- (b) Differentiate **2**

$$e^x(x^2 + 1)$$

(c) Differentiate

2

$$\frac{4x^2}{1-x}$$

Question 17 (5 marks)

There are twelve chocolates in a box. Four of the chocolates have mint centres and eight have caramel centres. Ali randomly selects two chocolates and eats them.

(a) Draw a tree diagram

2

(b) What is the probability that the two chocolates have mint centres?

1

(c) What is the probability that the two chocolates have the different centres?

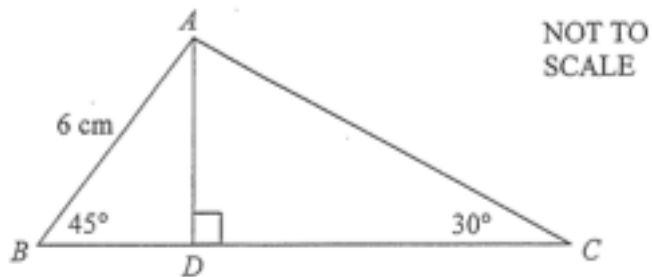
2

Question 18 (2 marks)

Sketch the graph of the function $f(x) = 1 - \log_e x$ showing the x intercept and the equation of the asymptote.

2

Question 19 (4 marks)



In the diagram, ABC is a triangle in which $\angle ABC = 45^\circ$, $\angle ACB = 30^\circ$ and $AB = 6\text{ cm}$. AD is the perpendicular from A to BC .

- (a) Find in simplest exact form the length of AD . 2

- (b) Find in simplest exact form the length of DC . 2

Question 20 (2 marks)

What is the domain of the function $f(x) = \frac{1}{x^2 - 4}$ 2

Question 21 (2 marks)

Use the formula $S = V_0(1 - r)^n$ to find correct to 2 decimal places the time (n years) **2**
it takes for an item with initial value $V_0 = \$2000$ to depreciate to the value $S = \$100$ at
an annual rate of depreciation $r = 10\%$.

Question 22 (2 marks)

(a) Solve x for the domain $0 \leq x \leq 2\pi$ **2**

$$2\sin x + \sqrt{3} = 0$$

Question 23 (3 marks)

Find the coordinates of the point where the gradient of the tangent is zero, on the curve

$$y = (x - 1)(x - 2)^3$$
 3

Question 24 (2 marks)

Use the formula $f'(x) = \frac{f(x+h)-f(x)}{h}$ to find $f'(x)$ when $f(x) = x^2 - 2x$.

2

Question 25 (3 marks)

Show that $\frac{1}{1+\sin x} + \frac{1}{1-\sin x} = 2\sec^2 x$

3

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Question 26 (3 marks)

- (a) Show that the function $f(x) = |x| - 1$ is even. **1**

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- (b) Sketch the graph of the curve $f(x) = |x| - 1$ showing the intercepts on the axes. **2**

Question 27 (4 marks)

- (a) Find the equation of the tangent to the curve $y = \frac{8}{3x-1}$ at the point where $x = 1$. **3**

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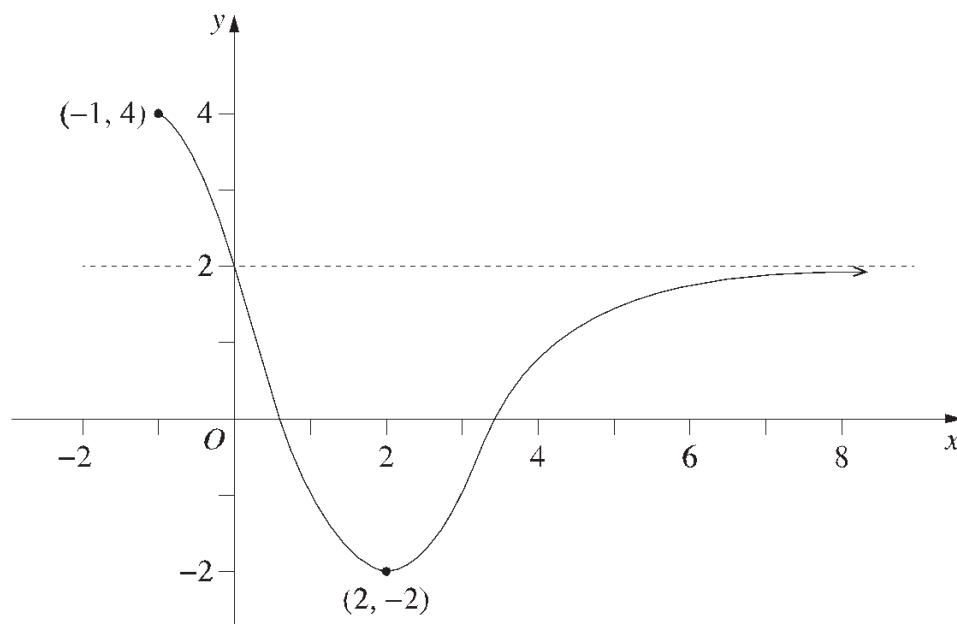
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(b) Find the angle of inclination of this tangent correct to the nearest degree. 1

Question 28 (2 marks)

2

The graph of a function $f(x)$ is shown. It has an asymptote at $y = 2$.



State the domain and range of $f(x)$

Question 29 (4 marks)

For a certain brand of medicine, the amount M present in the blood after t hours is given by

$$M = 3t^2 - t^3, \text{ for } 0 \leq t \leq 3$$

- (a) Find the average rate of change between $t = 0$ and $t = 3$ **2**

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- (b) Find the instantaneous rate of change $\frac{dM}{dt}$ at $t = 1$ **2**

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Question 30 (2 marks)

Solve for x $\log_2(x + 1) + \log_2(x - 1) = 3$.

2

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Question 31 (3 marks)

The energy, E Megajoules, radiated by an earthquake is given by the formula

$E = k \times 10^{1.5M}$ where M is the magnitude of the earthquake on the Richter scale

And k is a constant. An earthquake of magnitude 1 radiates 2 Megajoules of energy.

(a) Show that $\log_{10} E = \log_{10} 2 + 1.5(M - 1)$.

2

(b) Find the magnitude of an earthquake which radiates 2×10^6 Megajoules of energy.

1

End of paper

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

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Student Number: _____

Preliminary Mathematics Advanced

**Yearly Examination
September 2020**

Section I

Multiple-Choice Answer Sheet
Circle the answer of your choice.

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|-----------|---|---|---|---|
| 1. | A | B | C | D |
| 2. | A | B | C | D |
| 3. | A | B | C | D |

- | | | | | |
|-----|---|---|---|---|
| 4. | A | B | C | D |
| 5. | A | B | C | D |
| 6. | A | B | C | D |
| 7. | A | B | C | D |
| 8. | A | B | C | D |
| 9. | A | B | C | D |
| 10. | A | B | C | D |



KAMBALA

AS - MC & Q 11
PS - Q12 - 18
KH - Q19 = 25
SU - Q26 - 31

Student Number: _____

*Final with Solutions
7-9-20*

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Total marks – ~~70~~ 70

Section I 10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II 60 marks

- Attempt Questions 11 through 26
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt questions 1-10.

Allow about 15 minutes for this section.

Use the answer sheet for Questions 1 – 10.

1. What is the value of $\sqrt{48} - \sqrt{12}$?

(A) 2

(B) $\sqrt{12}$ (C) $\sqrt{24}$

(D) 6

$$4\sqrt{3} - 2\sqrt{3} = 2\sqrt{3} = \sqrt{12}$$

2. What are the x intercepts of the parabola $y = x^2 - 2x - 3$?

(A) -3 and -1

(B) -3 and 1

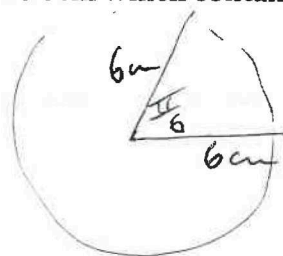
(C) -1 and 3

(D) 1 and 3

$$\begin{aligned} x^2 - 2x - 3 &= 0 \\ (x-3)(x+1) &= 0 \\ x &= 3 \text{ and } -1 \end{aligned}$$

3. What are the solutions of the equation $|2x + 1| - 3 = 0$?(A) $x = 2$ or $x = -1$ (B) $x = -2$ or $x = -1$ (C) $x = -2$ or $x = 1$ (D) $x = 2$ or $x = 1$

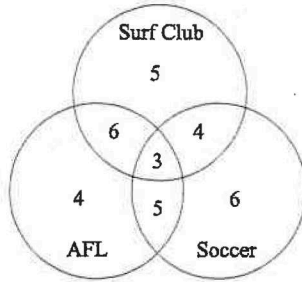
$$\begin{aligned} |2x+1| &= 3 \\ 2x+1 &= 3 \quad \text{or} \quad 2x+1 = -3 \\ 2x &= 2 \quad \quad \quad 2x = -4 \\ x &= 1 \quad \quad \quad x = -2 \end{aligned}$$

4. What is the perimeter of a sector of a circle of radius 6cm which contains a sector angle of $\frac{\pi}{6}$ radians at the centre of the circle?(A) π cm(B) 3π cm(C) $(12 + \pi)$ cm(D) $(12 + 3\pi)$ cm

$$L = r\theta = 6 \times \frac{\pi}{6} = \pi \text{ cm}$$

$$\begin{aligned} P &= 6 + 6 + \pi \\ &= 12 + \pi \end{aligned}$$

5. In a classroom, students are asked what sports club they are members of and the results are shown in the Venn diagram.



A student who is a member of the soccer club is chosen at random. What is the probability that he/she is also a member of a surf club?

(A) $\frac{2}{5}$

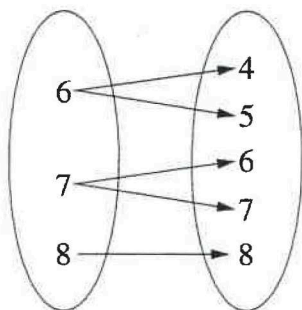
(B) $\frac{4}{11}$

(C) $\frac{2}{9}$

(D) $\frac{7}{18}$

$\frac{7}{18}$ surf & soccer
soccer

6. What type of relation is this?



(A) Many-to-many

(B) One-to-many

(C) One-to-one

(D) Many-to-one

7. If $y = \sqrt{2x + 1}$ what is the expression for $\frac{dy}{dx}$?

(A) $\frac{dy}{dx} = 2\sqrt{2x + 1}$

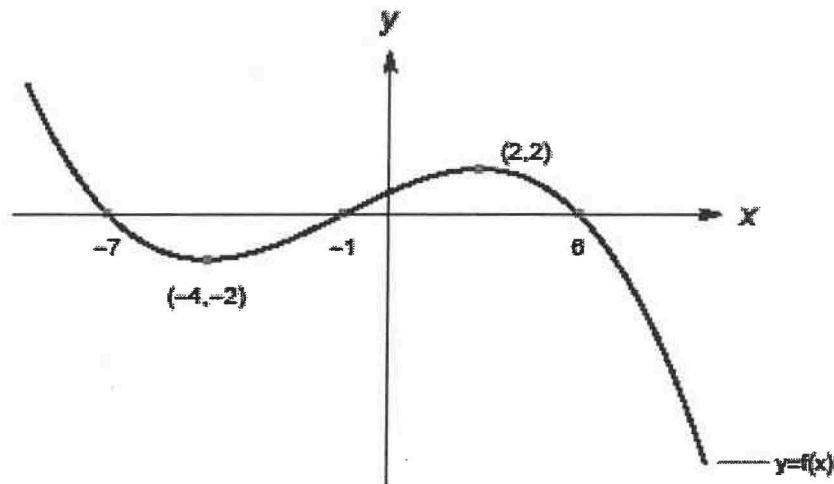
(B) $\frac{dy}{dx} = \frac{1}{2\sqrt{2x+1}}$

(C) $\frac{dy}{dx} = \frac{1}{\sqrt{2x+1}}$

(D) $\frac{dy}{dx} = \frac{2}{\sqrt{2x+1}}$

$$\frac{dy}{dx} = \frac{1}{2} \times 2(2x+1)^{-\frac{1}{2}}$$

8.



For the graph $y = f(x)$ shown above, $f'(x)$ is negative when

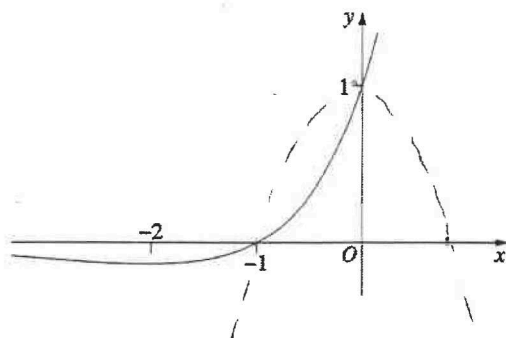
(A) $-7 < x < -1$ or $x > 6$

(B) $x < -4$ or $x > 2$

(C) $-4 < x < 2$

(D) $-2 < x < 2$

9. The diagram shows the graph of $y = e^x(1 + x)$ below.



How many solutions to $y = e^x(1 + x) = 1 - x^2$?

(A) 0

(B) 1

(C) 2

(D) 3

$$1 - x^2 = 0$$

$$x^2 = 1$$

$$x = \pm 1$$

10. If $a = e^x$ what is the correct expression for $\log_e \frac{1}{a^2}$?

(A) $-x^2$

(B) $-2x$

(C) $2x$

(D) x^2

$$\begin{aligned} \log_e \left(\frac{1}{e^x} \right)^2 &= \log_e e^{-2x} \\ &= -2x \end{aligned}$$

Section II

60 marks

Attempt all questions.

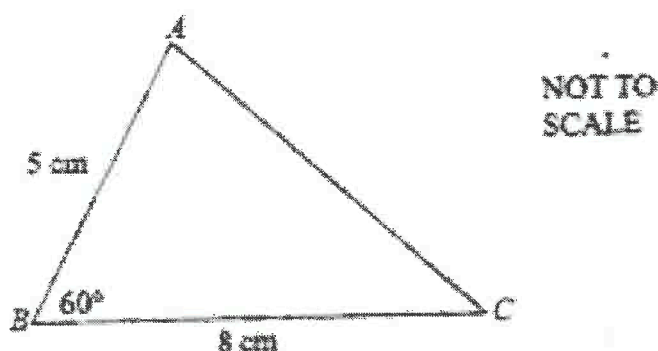
Allow about 1 hour and 45 minutes for this section.

Answer the questions in the spaces provided.

Your responses should include relevant mathematical reasoning and/or calculations.

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Question 11 (3 marks)



ABC is a triangle in which $AB = 5\text{cm}$, $BC = 8\text{cm}$ and $\angle ABC = 60^\circ$

(a) Find the length of AC.

2

$$\begin{aligned} b^2 &= 5^2 + 8^2 - 2(5)(8)\cos 60^\circ \\ &= 25 + 64 - 80 \times \frac{1}{2} \\ &= 49 \\ b &= 7 \\ AC &= 7 \text{ cm} \end{aligned}$$

(b) Find in simplest exact form the area of $\triangle ABC$

1

$$\begin{aligned} \text{Area} &= \frac{1}{2}(5)(8)\sin 60^\circ \\ &= 20 \times \frac{\sqrt{3}}{2} \\ &= 10\sqrt{3} \text{ cm}^2 \end{aligned}$$

Question 12 (2 marks)(a) Solve for x

$$\frac{x-2}{4} + \frac{7-x}{6} = 1$$

$$3(x-2) + 2(7-x) = 12$$

$$3x - 6 + 14 - 2x = 12$$

$$x + 8 = 12$$

$$x = 4$$

2

Question 13 (2 marks)A(-2,1), B(2,-1) and C(4,3) are the vertices of a triangle. Show that $AB \perp BC$. 2

$$m_1 = \text{Gradient } AB = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1 - 1}{2 - (-2)} = \frac{-2}{4} = -\frac{1}{2}$$

$$m_2 = \text{Gradient } BC = \frac{3 - (-1)}{4 - 2} = \frac{4}{2} = 2$$

$$m_1 m_2 = -\frac{1}{2} \times 2 = -1 \therefore AB \perp BC$$

$$P(\text{even}) = \frac{1}{2}$$

$$P(\text{multiple of 3}) = \frac{1}{3}$$

Question 14 (2 marks)

A fair die is rolled once. A is the event the score is even and B is the event a multiple of 3. Explain whether events A and B are independent. 2

$$A = \{2, 4, 6\} \quad P(A) = \frac{3}{6} = \frac{1}{2} \checkmark$$

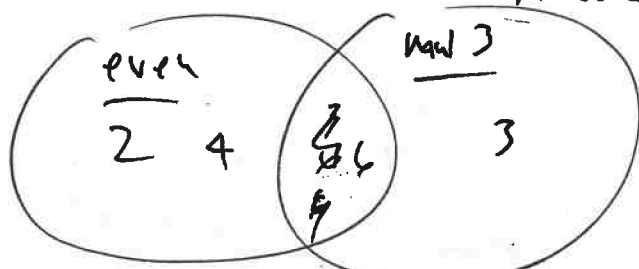
$$B = \{3, 6\} \quad P(B) = \frac{2}{6} = \frac{1}{3}$$

$$A \cap B = \{6\} \quad P(A \cap B) = \frac{1}{6}$$

$$P(A \cap B) = P(A) \times P(B)$$

$$\frac{1}{6} = \frac{1}{2} \times \frac{1}{3}$$

Yes A and B are independent events



$$P(A \text{ even} | \text{multiple of 3}) = \frac{1}{2}$$

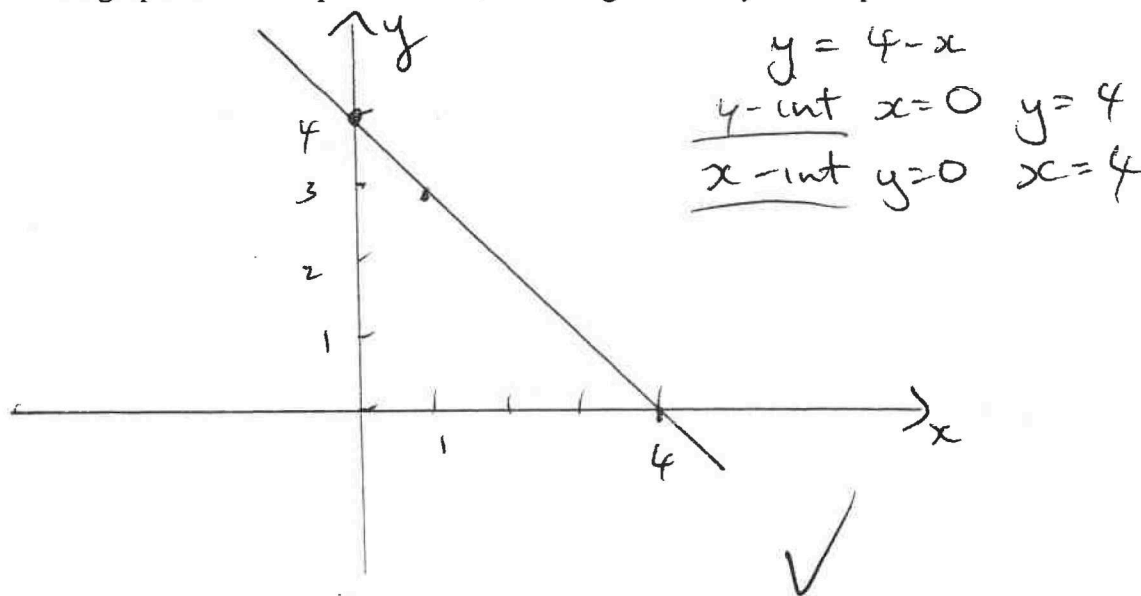
$$P(\text{multiple of 3} | \text{even}) = \frac{1}{3}$$

Question 15 (3 marks)

- (a) If
- $f(x) = 4 - x^2$
- and
- $g(x) = \sqrt{x}$
- find the composite function
- $f(g(x))$
- .
- 1

$$f(g(x)) = 4 - (\sqrt{x})^2 = 4 - x$$

- (b) Sketch the graph of this composite function showing the
- x
- and
- y
- intercepts.
- 2

**Question 16** (5 marks)

- (a) Differentiate
- 1

$$x^4 + 2x^2 + 10 \quad \frac{dy}{dx} = 4x^3 + 4x$$

- (b) Differentiate
- 2

$$e^x(x^2 + 1) \quad \frac{dy}{dx} = e^x \times 2x \times (x^2 + 1) + e^x(x^2 + 1)^2$$

$$= e^x(x^2 + 1)(4x + x^2 + 1)$$

$$e^x(x^2 + 1) \quad \frac{dy}{dx} = e^x \times 2x + e^x \times (x^2 + 1)$$

$$= 2xe^x + e^x(x^2 + 1)$$

(c) Differentiate

2

$$\frac{4x^2}{1-x} = \frac{u}{v} \quad \frac{dy}{dx} = \frac{(1-x) \times 8x - 4x^2(-1)}{(1-x)^2}$$

$$= \frac{8x - 8x^2 + 4x^2}{(1-x)^2}$$

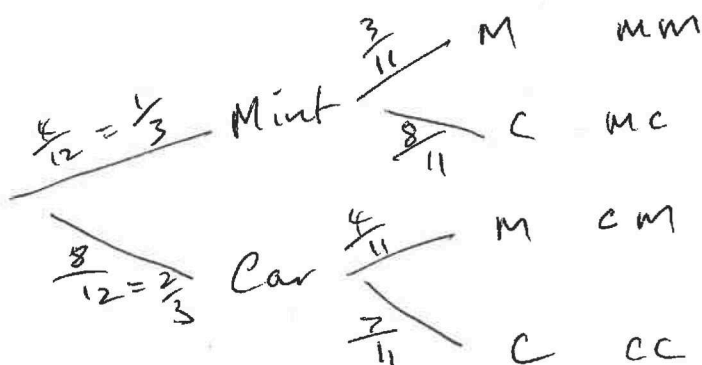
$$= \frac{8x - 4x^2}{(1-x)^2}$$

Question 17 (5 marks)

There are twelve chocolates in a box. Four of the chocolates have mint centres and ^{eight} ~~four~~ have caramel centres. Ali randomly selects two chocolates and eats them.

(a) Draw a tree diagram

2



(b) What is the probability that the two chocolates have mint centres?

1

$$P(MM) = \frac{1}{3} \times \frac{3}{11} = \frac{1}{11}$$

(c) What is the probability that the two chocolates have the different centres?

2

$$P(MC) + P(CM) = \frac{1}{3} \times \frac{8}{11} + \frac{2}{3} \times \frac{4}{11}$$

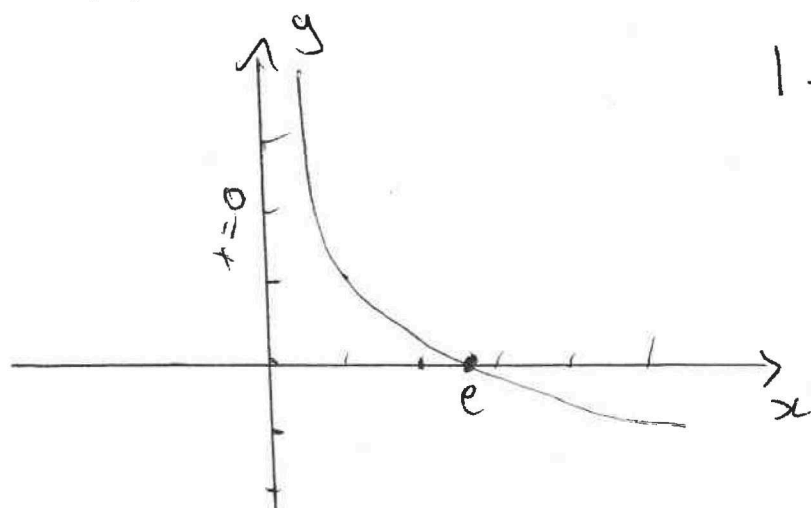
$$= \frac{8}{33} + \frac{8}{33}$$

$$= \frac{16}{33}$$

Question 18 (2 marks)

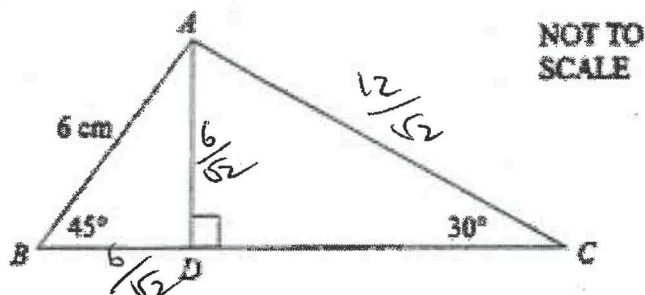
Sketch the graph of the function $f(x) = 1 - \log_e x$ showing the x intercept and the equation of the asymptote.

2



$$\begin{aligned} 1 - \log_e x &= 0 \\ \log_e x &= 1 \\ e^1 &= x \\ x &\approx 2.7 \dots \end{aligned}$$

Question 19 (4 marks)



In the diagram, ABC is a triangle in which $\angle ABC = 45^\circ$, $\angle ACB = 30^\circ$ and $AB = 6$ cm .
 ADS is the perpendicular from A to BC .

(a) Find in simplest exact form the length of AD.

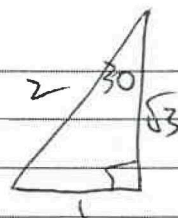
2

$$AD = \frac{6}{\sqrt{2}} \times \frac{\sqrt{2}}{2} = \frac{6\sqrt{2}}{2} = 3\sqrt{2}$$

(b) Find in simplest exact form the length of DC.

2

$$DC = 3\sqrt{2} \times \sqrt{3} = 3\sqrt{6}$$



Question 20 (2 marks)

What is the domain of the function $f(x) = \frac{1}{x^2-4}$

2

All real except $x = \pm 2$

Question 21 (2 marks)

Use the formula $S = V_0(1 - r)^n$ to find correct to 2 decimal places the time (n years) it takes for an item with initial value $V_0 = \$2000$ to depreciate to the value $S = \$100$ at an annual rate of depreciation $r = 10\%$. 2

$$100 = 2000(1 - 0.1)^n$$

$$\frac{100}{2000} = 0.9^n$$

$$\log \frac{1}{20} = n \log 0.9$$

$$\frac{\log \frac{1}{20}}{\log 0.9} = n$$

$$n = 28.43 \text{ years}$$

Question 22 (2 marks)

(a) Solve x for the domain $0 \leq x \leq 2\pi$ 2

$$2 \sin x + \sqrt{3} = 0$$

$$2 \sin x = -\sqrt{3}$$

$$\sin x = -\frac{\sqrt{3}}{2}$$

$$x = \pi + \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$$

$$= \frac{4\pi}{3}, \frac{5\pi}{3}$$

ref angle = $\frac{\pi}{3}$ or 60°

S	A
T	F
✓	✓

Question 23 (3 marks)

Find the coordinates of the point where the gradient of the tangent is zero, on the curve

$$y = (x - 1)(x - 2)^3$$

$$\frac{dy}{dx} = (x - 1) \times 3(x - 2)^2 + (x - 2)^3 \times 1$$

$$= (x - 2)^2 (3(x - 1) + (x - 2))$$

$$= (x - 2)^2 (3x - 3 + x - 2)$$

$$= (x - 2)^2 (4x - 5) = 0$$

$$4x - 5 = 0$$

$$x = 2 \text{ or } x = \frac{5}{4}$$

Question 24 (2 marks)Use the formula $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ when $f(x) = x^2 - 2x$.

2

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(x+h)^2 - 2(x+h) - (x^2 - 2x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 2x - 2h - x^2 + 2x}{h} \\
 &= \lim_{h \rightarrow 0} \frac{h(2x + h - 2)}{h} \\
 &= \lim_{h \rightarrow 0} 2x + h - 2 \quad f'(x) = 2x - 2
 \end{aligned}$$

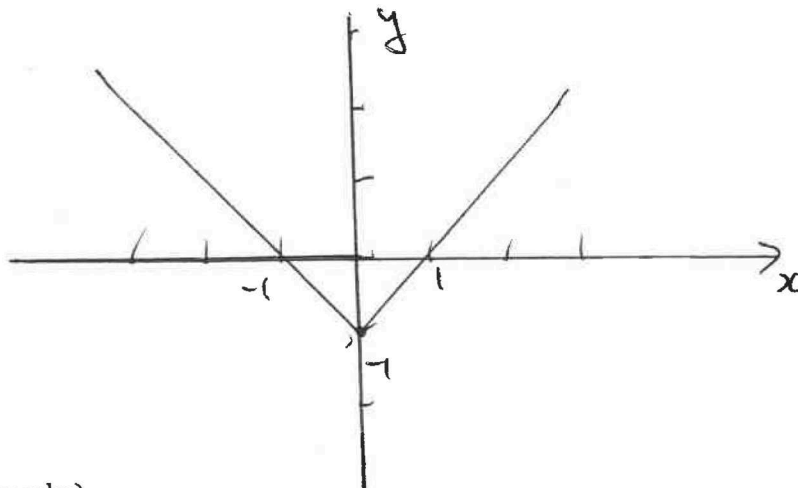
Question 25 (3 marks)Show that $\frac{1}{1+\sin x} + \frac{1}{1-\sin x} = 2\sec^2 x$

3

$$\begin{aligned}
 \text{LHS} &= \frac{1 - \sin x + 1 + \sin x}{(1 + \sin x)(1 - \sin x)} \\
 &= \frac{2}{1 - \sin^2 x} \\
 &= \frac{2}{\cos^2 x} \\
 &= 2 \sec^2 x \\
 &= \text{RHS}
 \end{aligned}$$

Question 26 (3 marks)(a) Show that the function $f(x) = |x| - 1$ is even. 1

Even $f(-x) = f(x)$ $f(-x) = |-x| - 1$
 $= |x| - 1$
 $\therefore$ even function $= f(x)$

(b) Sketch the graph of the curve $f(x) = |x| - 1$ showing the intercepts on the axes. 2**Question 27** (4 marks)(a) Find the equation of the tangent to the curve $y = \frac{8}{3x-1}$ at the point where $x = 1$. 3

$y = 8(3x-1)^{-1}$

$$\frac{dy}{dx} = \frac{(3x-1) \times 0 - 8 \times 3(3x-1)^{-2}}{(3x-1)^2} \quad \frac{dy}{dx} = \frac{-24}{(3x-1)^2}$$

$$= \frac{-24}{(3x-1)^2}$$

$$f'(1) = \frac{-24}{4} = -6$$

$x=1 \quad y = \frac{8}{3-1} = \frac{8}{2} = 4 \quad (1, 4)$

$$y - 4 = -6(x - 1)$$

$$y - 4 = -6x + 6$$

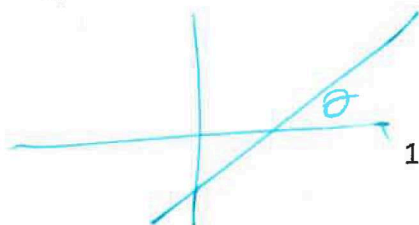
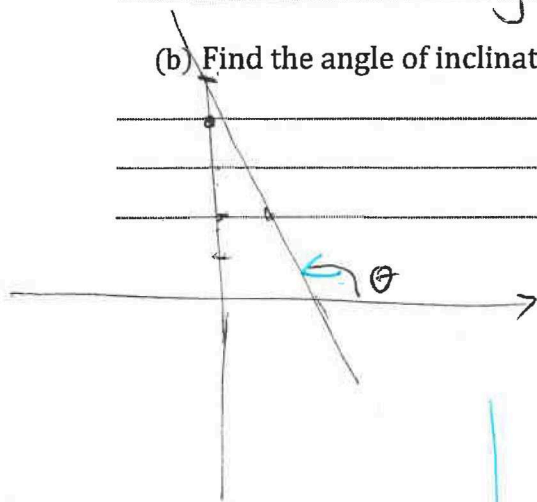
$$y = -6x + 10$$

(b) Find the angle of inclination of this tangent correct to the nearest degree. 1

$$\tan \theta = \frac{-6}{1} = -6 \Rightarrow \theta = 99.46^\circ$$

$$\theta = 99^\circ$$

$$\theta = 99^\circ$$

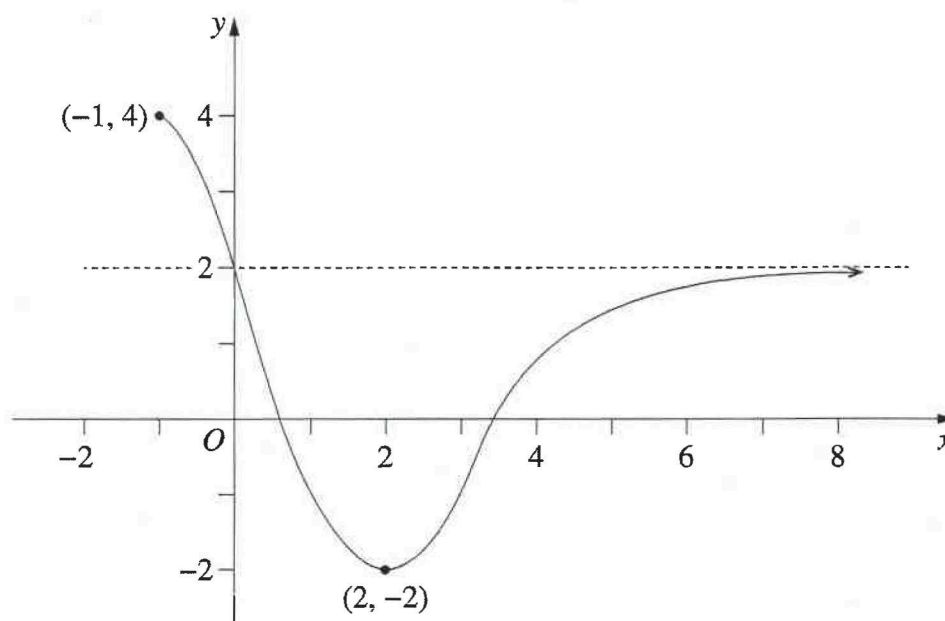


$$\tan \theta = \frac{y}{x} = \frac{\text{rise}}{\text{run}} = \text{gradient}$$

Question 28 (2 marks)

2

The graph of a function $f(x)$ is shown. It has an asymptote at $y = 2$.



State the domain and range of $f(x)$

$$D : x \geq -1$$

$$R : -2 \leq y \leq 4$$

Question 29 (4 marks)

For a certain brand of medicine, the amount M present in the blood after t hours is given by

$$M = 3t^2 - t^3, \text{ for } 0 \leq t \leq 3$$

- (a) Find the average rate of change between $t = 0$ and $t = 3$

$$f(3) = 27 - 27 = 0$$

$$f(0) = 0$$

$$\text{Average rate of change} = \frac{f(3) - f(0)}{3 - 0} = \frac{0 - 0}{3} = 0$$

- (b) Find the instantaneous rate of change $\frac{dM}{dt}$ at $t = 1$

2

$$\frac{dM}{dt} = 6t - 3t^2$$

$$f'(1) = 6 - 3 = 3$$

Question 30 (2 marks)

2

Solve for x $\log_2(x+1) + \log_2(x-1) = 3$.

$$\log_2(x^2 - 1) = 3$$

$$2^3 = x^2 - 1$$

$$8 + 1 = x^2$$

$$x^2 = 9 \quad x > 1$$

$$x = +3$$

Question 31 (3 marks)

The energy, E Megajoules, radiated by an earthquake is given by the formula

$E = k \times 10^{1.5M}$ where M is the magnitude of the earthquake on the Richter scale

And k is a constant. An earthquake of magnitude 1 radiates 2 Megajoules of energy.

- (a) Show that $\log_{10} E = \log_{10} 2 + 1.5(M - 1)$.

2

$$E = k \times 10^{1.5M}$$

$$M = 1 \quad E = 2$$

$$\frac{E}{k} = 10^{1.5M}$$

$$2 = k \times 10^{1.5}$$

$$\log_{10} \frac{E}{k} = 1.5M$$

$$k = \frac{2}{10^{1.5}} = 0.063$$

$$M = \frac{1}{1.5} \log_{10} \frac{E}{k}$$

$$M = \frac{1}{1.5} (\log_{10} E - \log_{10} k)$$

$$1.5M = \log_{10} E - \log_{10} \frac{2}{10^{1.5}}$$

$$= \log_{10} E - (\log_{10} 2 - \log_{10} 10^{1.5})$$

$$1.5M = \log_{10} E - \log_{10} 2 + \log_{10} 10^{1.5}$$

- (b) Find the magnitude of an earthquake which radiates 2×10^6 Megajoules of energy.

1

$$\log_{10} E = \log_{10} 2 + 1.5M - 1.5$$

$$\log_{10} (2 \times 10^6) = \log_{10} 2 + 1.5M - 1.5$$

$$6 = 1.5M$$

$$M = 4$$

End of paper

$$\log_{10} E = \log_{10} 2 + 1.5M - 1.5$$

$$\log_{10} E = \log_{10} 2 + 1.5(M - 1)$$

Section II extra writing space

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Student Number: _____

Preliminary Mathematics Advanced

Yearly Examination
September 2020

Section I

Multiple-Choice Answer Sheet
Circle the answer of your choice.

- | | | | | |
|-----|---|----------|----------|----------|
| 1. | A | <u>B</u> | C | D |
| 2. | A | B | <u>C</u> | D |
| 3. | A | B | <u>C</u> | D |
| 4. | A | B | <u>C</u> | D |
| 5. | A | B | C | <u>D</u> |
| 6. | A | <u>B</u> | C | D |
| 7. | A | B | <u>C</u> | D |
| 8. | A | <u>B</u> | C | D |
| 9. | A | B | <u>C</u> | D |
| 10. | A | <u>B</u> | C | D |

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